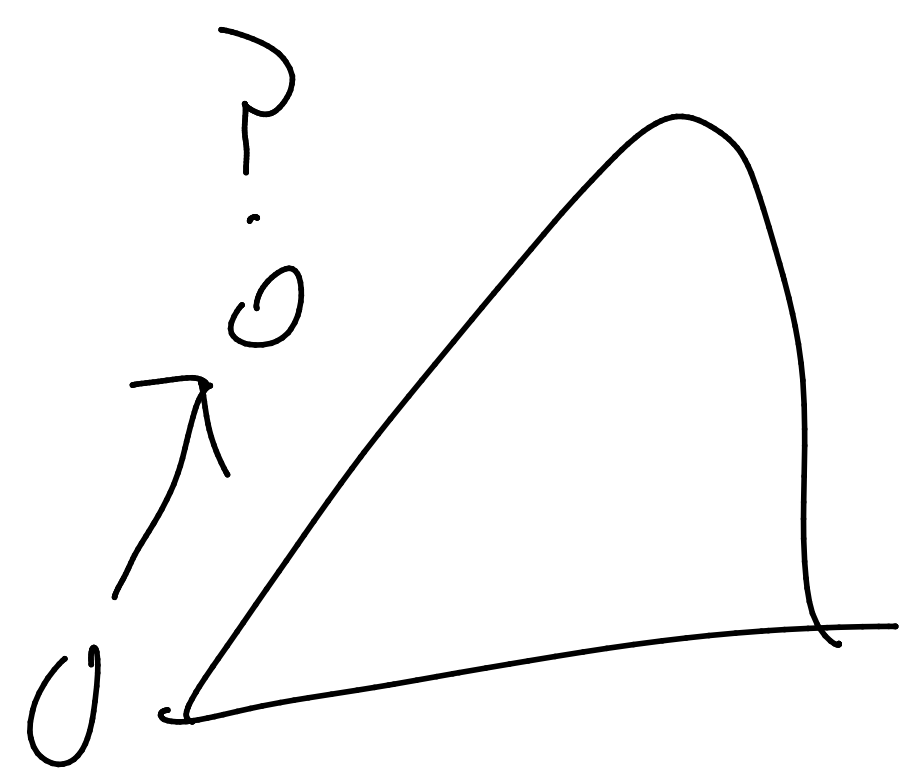


1a)



$$h_0 = 0$$

$$mgh_f = \frac{1}{2}mv_0^2 + \frac{1}{2}I\omega^2$$

$$I_0 h_f = \frac{7}{10}v_0^2$$

$$h_f = 0.74 \text{ m}$$

2)

a)

(cylinder $I = MR^2$

ball $I = \frac{2}{5}MR^2$

as the ball has less rotational inertia,
it will travel faster and reach the bottom
first

b)

as the ball has a lower $acc.$, it ought to
start as a lower point, but both objects
ought to have same final velocity.

c)

Sphere

$$mgh_0 = \frac{1}{2}mv^2 + \frac{2}{5}MR^2\omega^2$$

$$gh_0 = \frac{7}{10}v^2$$

(cylinder

$$mgh_0 = \frac{1}{2}mv^2 + MR^2\omega^2$$

$$gh_0 =$$